



OpenShift Virtualization DR Battlecard

Technical presentation

How to use this presentation

- ▶ Red Hat's response to how create a DR solution for OpenShift Virtualization is to use a partner as we don't offer this capability.
- ▶ One cannot always have a partner in the room to discuss this topic
- ▶ The intent of these slides is to help getting the conversation started with the customer by having a high-level idea of what each partner can do.
- ▶

Agenda

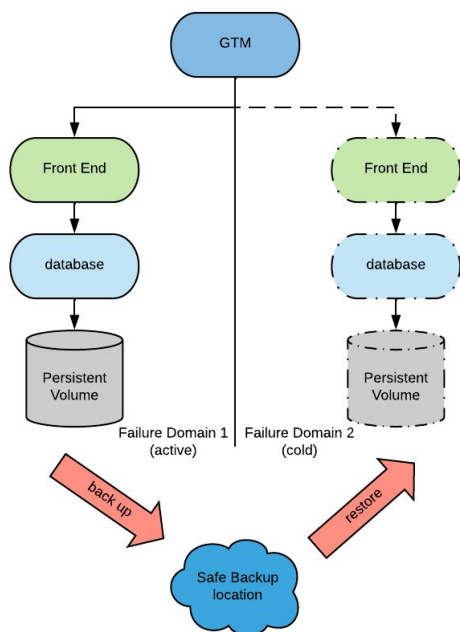
- ▶ [OpenShift Virtualization DR approaches](#)
- ▶ SAN Vendors
 - [Dell](#)
 - [HPE](#)
 - [Infinidat](#)
 - [Hitachi](#)
 - [NetApp](#)
 - [Pure](#)
 - [IBM Storage Virtualize on FlashSystem Family](#)
- ▶ SDS Vendors
 - [Portworx](#)
 - [ODF](#) ([see red hat's docs](#))
- ▶ Backup/Restore Vendors
 - [Trilio](#)

OpenShift Virtualization DR Architectural Approaches

High-level DR Approaches

Failure Domain is either a data center or a cloud region

Active/Passive



Backup / Restore

RPO $\approx$ frequency of backup
RTO $\approx$ hours

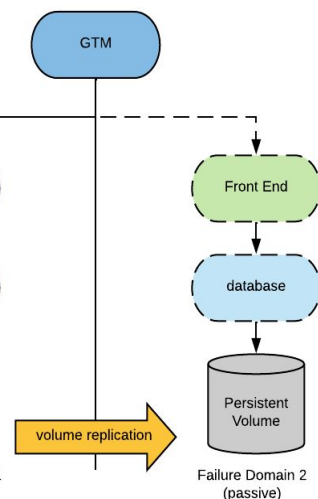
Volume Replication

Asynchronous:

- RPO $\approx$ unbounded (typically ms)
- RTO $\approx$ typically minutes *

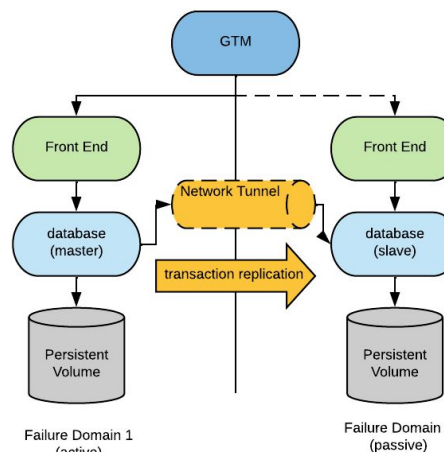
Synchronous:

- RPO = 0
- RTO $\approx$ typically minutes*

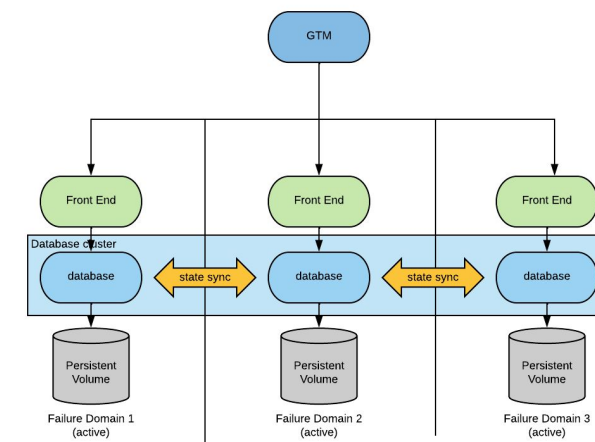


Transaction Replication

RPO = 0
RTO $\approx$ minutes



Active/Active



Distributed Stateful Workloads

Strong Consistency:

- RPO = 0
- RTO $\approx$ seconds

Eventual Consistency (**)

- RPO $\approx$ 0
- RTO $\approx$ seconds

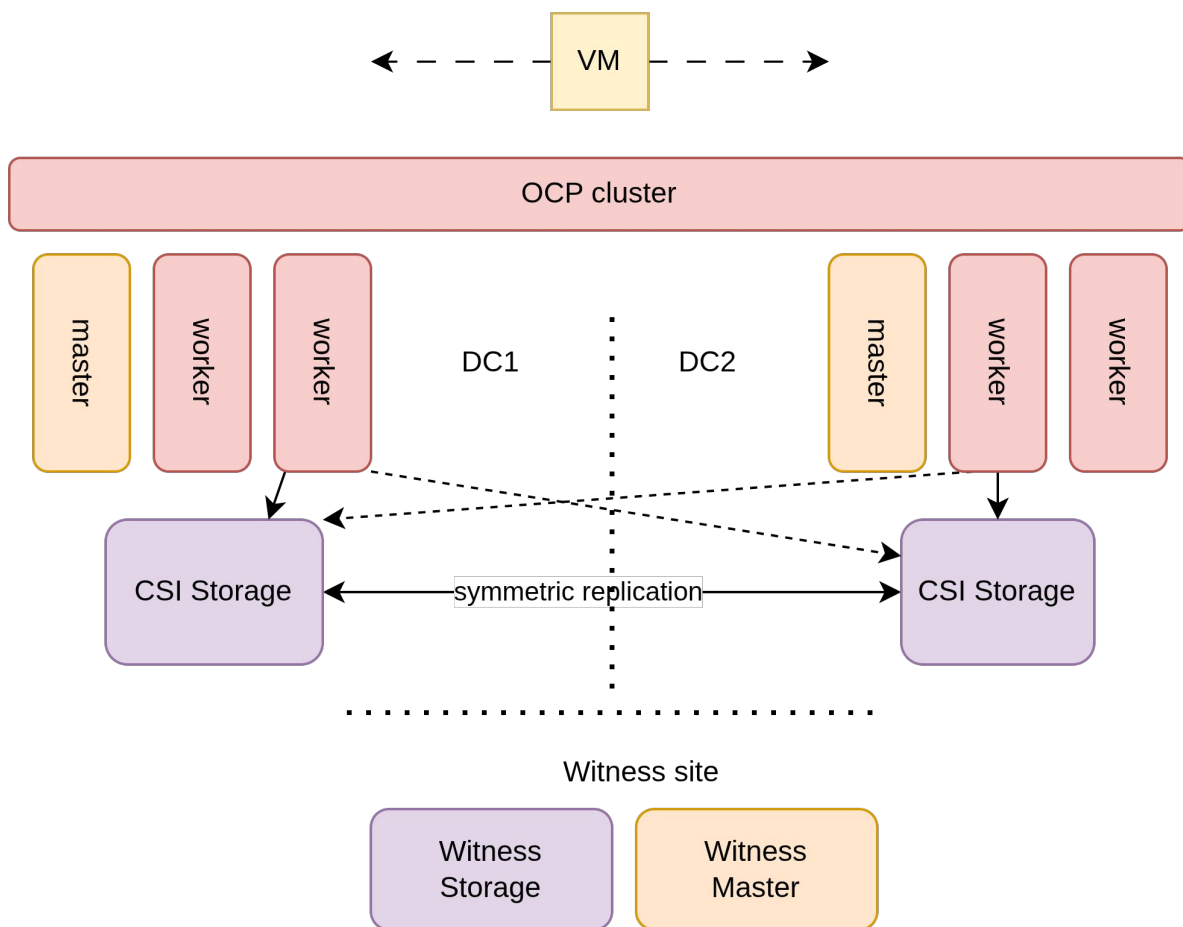
(*) a database can take an unbounded amount of time to recover from a crash consistent volume

(**) eventual consistency does not imply eventual correctness



Symmetric replication

A.k.a MetroDR -- Active/Active -- Latency < 5-10ms between DC1 and DC2



- OCP and Storage are stretched across DC1 and DC2
- A witness site is required for both OCP and storage
- Very little latency between DC1 and DC2, more latency is usually tolerated between the witness site and the main DCs
- In case of a disaster VMs will restart in the healthy DC.
- In case the storage fails, the CSI driver should implement the failover of the multipath connection.
- L2 VLANs for VMs must exist in both DCs.
- Labels and affinity rules might be used to have VMs prefer one DC or the other.
- CAUTION: a DR event may cause a VM restart storm, verify that storage and OCP control plane do not get overwhelmed.
- CAUTION: no restart order can be enforced (typically needed for DR plans), pod priority or node labels can be used partially control this behavior.

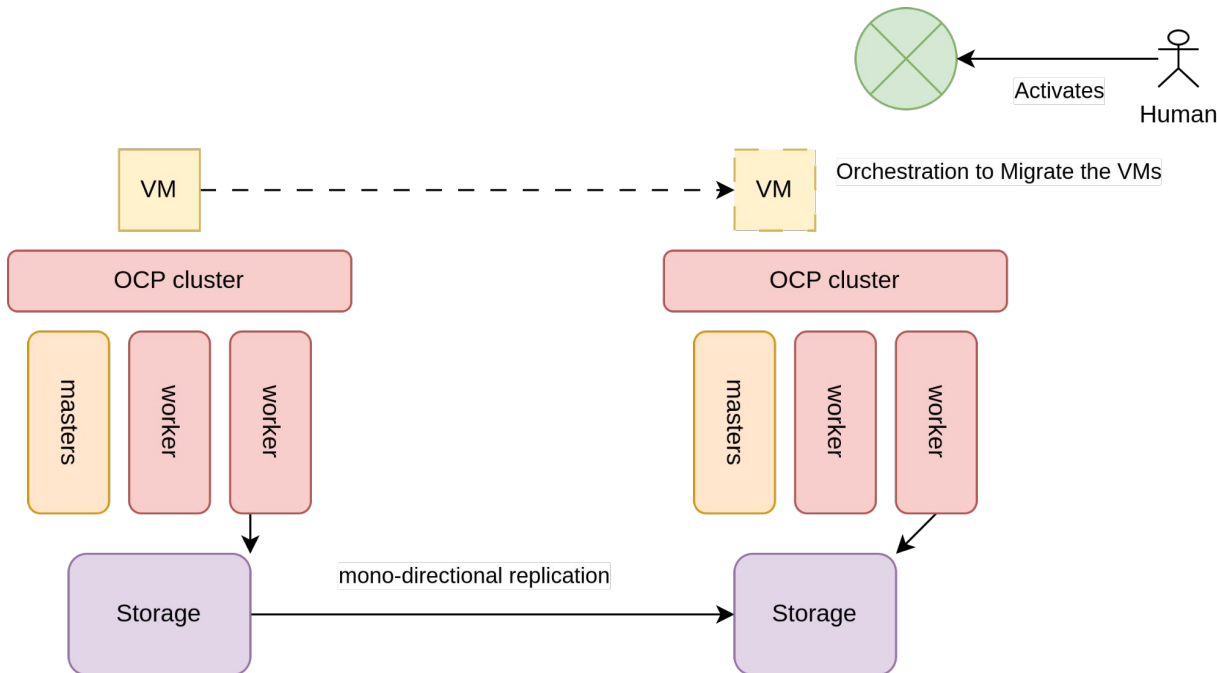
Note: some schools of thought do not consider this DR because the OCP control plane is a failure domain and if it breaks you lose the cluster

Note: Witness Master here means a normal master deployed in the witness site

Note: ensure to have at least 50% of available capacity in either side to support the VMs swinging during a DR event.

Mono-directional replication

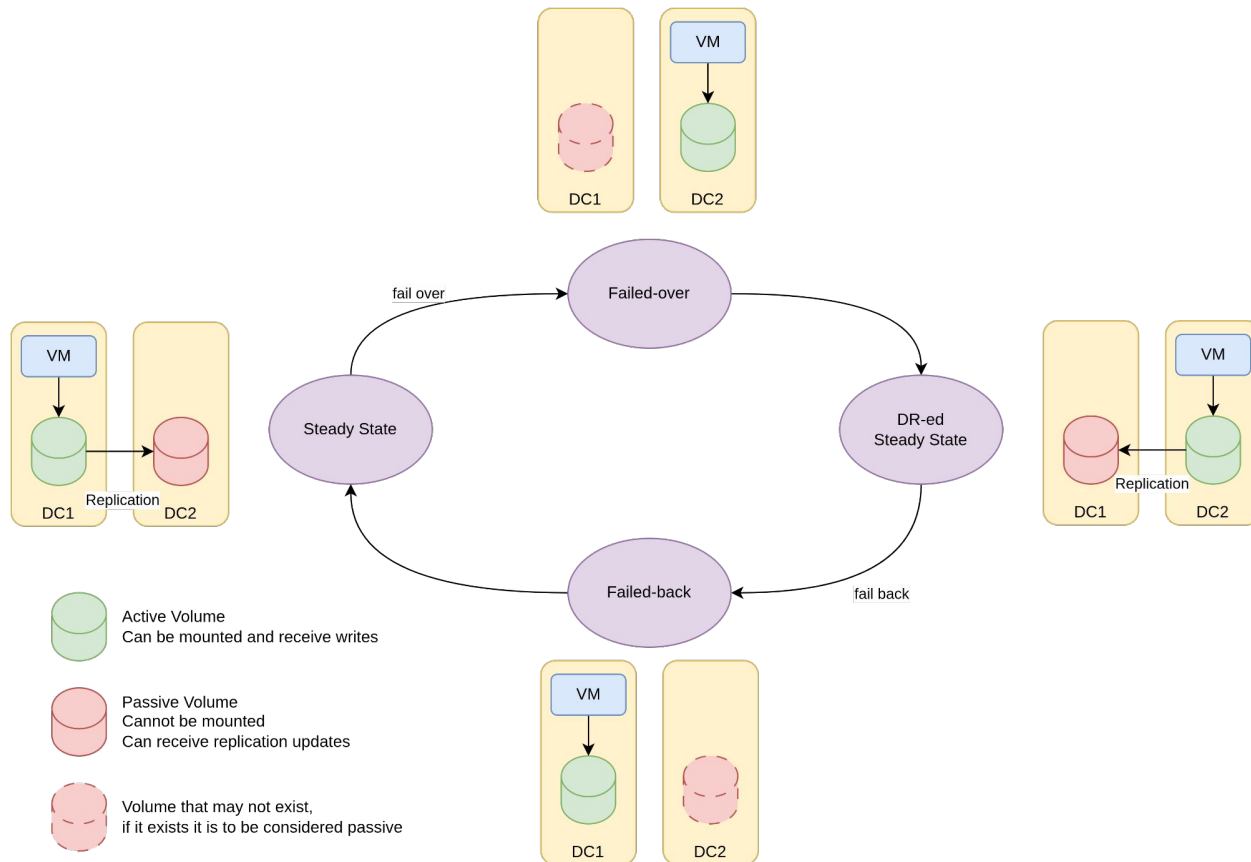
A.k.a Regional DR -- Active/Passive -- Latency > 10ms between DC1 and DC2



- VM Volumes are replicated to the other site (either sync or async replication)
- DR must be triggered externally, typically by a human.
- The DR process requires the following components:

Component	Description
Volume Group support	Ability to consistently replicate a group of volumes
Volume Replication Status management	Manage the direction of the replication and whether replication is active or not.
VM Management	Ability to start/stop VMs in the correct DC, attaching them to the same Volumes
Recovery orchestration	Process that restarts all the VMs in the healthy site in case of a disaster, throttling the VM restarts and managing dependencies,

Conceptual state diagram of a DR orchestration



State transitions should be applicable to individual VMs or groups of VMs in a single transaction. This would allow to support the use cases of:

1. Individual application failover
2. Failover of all the application in a tier
3. Manage throttling and precedences

Vendors expose the capabilities to create this orchestration in very different way. There is not today a storage agnostic way of writing this orchestration.

There are many ways of writing this orchestration. Ansible and gitops should be our go-tos

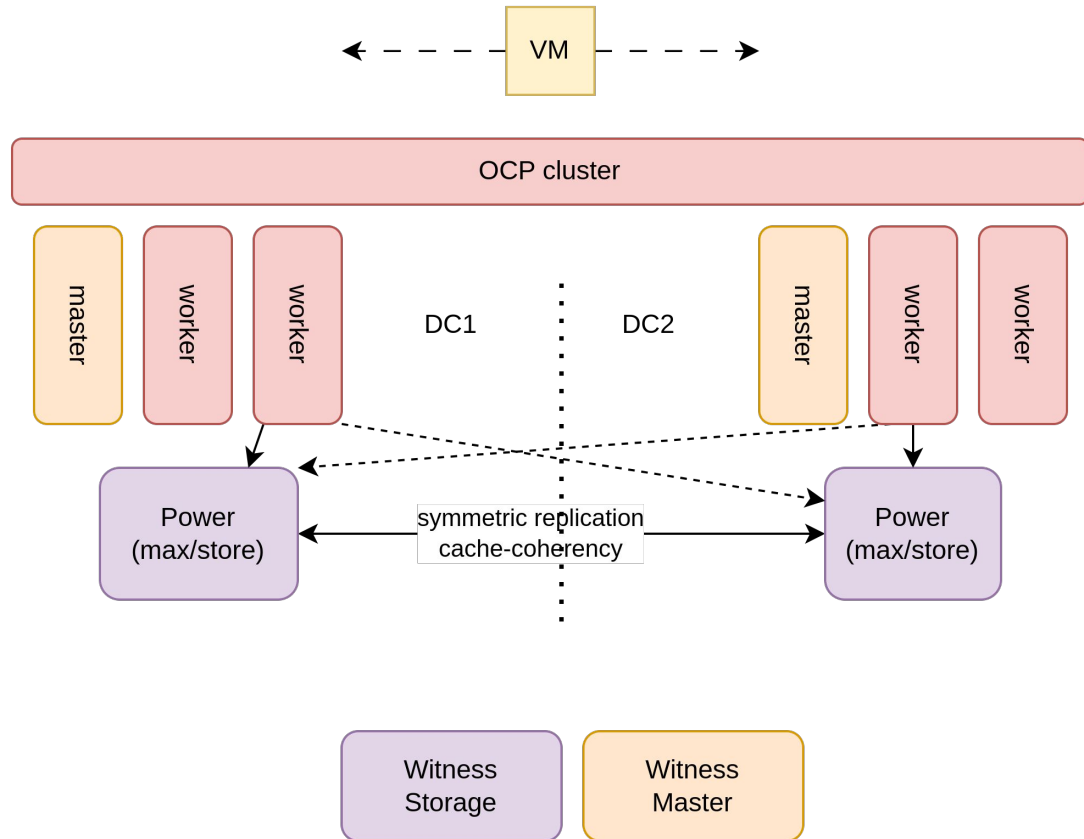
Dell

Dell Storage Models

Model Family	Positioning	CSI Drivers	Supports Metro DR	Supports Regional DR
PowerMax	High-end SAN	CSI	yes	yes
PowerStore	Mid-tier SAN	CSI	yes	yes
PowerFlex	SDS	CSI	no	yes
PowerScale	File storage	CSI	no	yes

Metro DR

RTO	~5 minutes
RPO	0

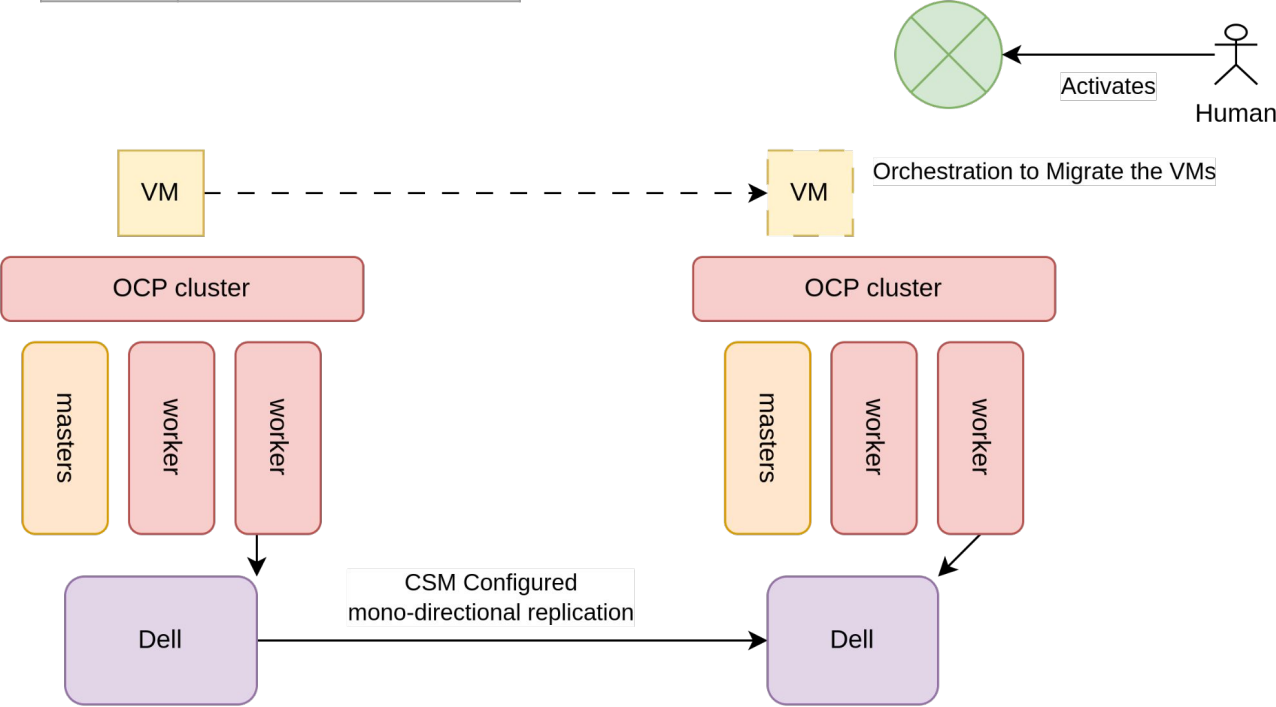


Synchronous replication for Dell Storage Array is based on **Cache Coherency** and documented here [PowerMax Metro](#), [PowerStore Metro](#). This architecture requires uniform topology for the LUNs.

CSI driver understands the **uniform** topology, including setting up preferred/secondary multipath LUNs. *multipathd.conf* for this setup is provided by Dell.

Regional DR

RTO	N/A
RPO	Min 0 when using sync



Component	Description
Volume Group Support	<u>yes</u>
Volume Replication Status management	Defined by <u>CSM extension</u> , replication status managed with <u>repctl</u> . Sync and async replication methods are supported.
VM Management	Not provided
Recovery orchestration	Not provided

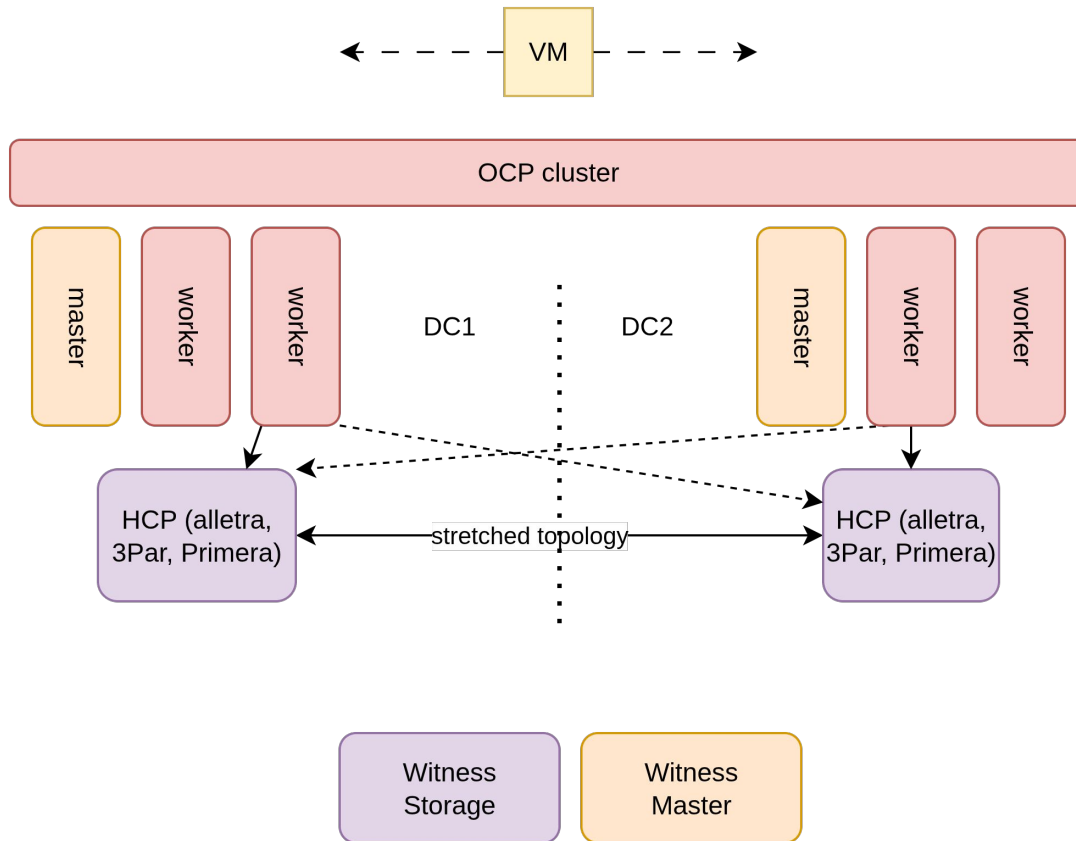
HPE

HPE Storage Models

Model Family	Positioning	CSI Drivers	Supports Metro DR	Supports Regional DR
Nimble	Mid-tier SAN (being phased out)	<u>CSI</u>	no	No (see slides for details)
3Par -> Primera -> Alletra	High-end to mid-tier SAN	<u>CSI</u> <u>See also</u>	<u>Yes</u> for HPE Alletra Storage MP B10000, Alletra 9000, Primera and 3PAR	No (see slides for details)
XP[7 8]	??	<u>CSI</u>		Yes. <u>via proprietary extension</u> .

Metro DR

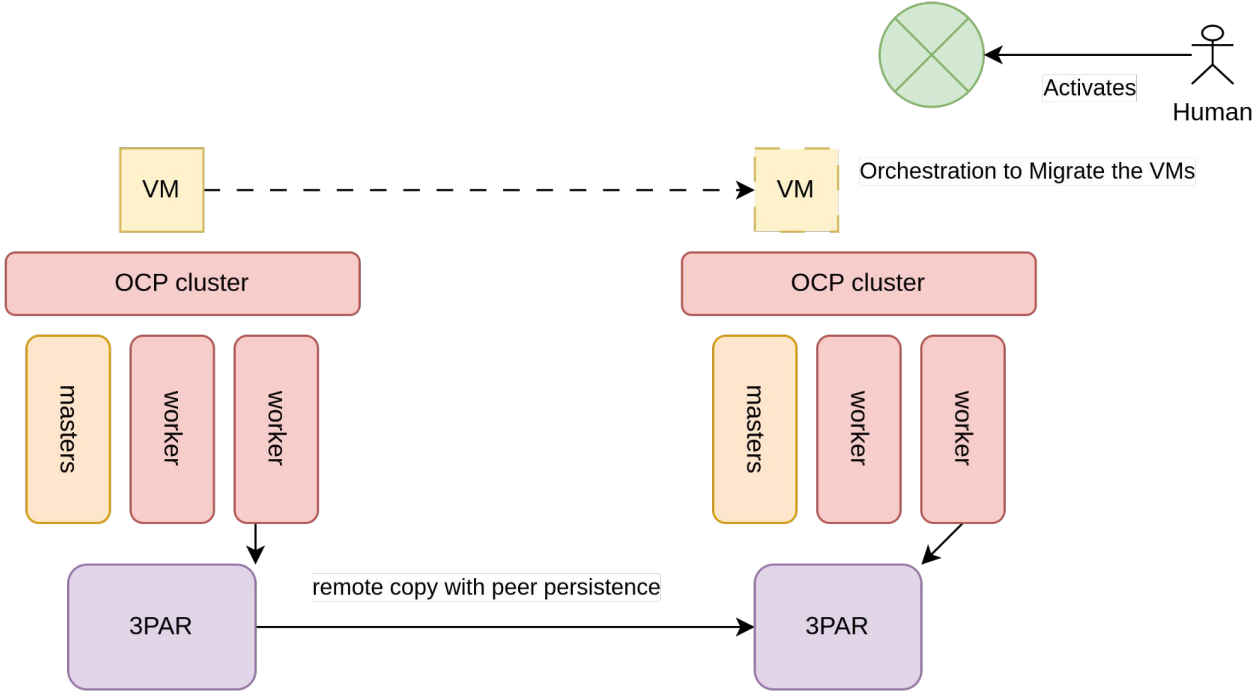
RTO	~5 minutes
RPO	0



HPE has two different [active/active synchronization mechanisms](#): Active Peer Persistence (APP) and Classic Peer Persistence (CPP). From the docs both seem suitable for the metro DR setup

Regional DR [3Par, Primera, Alletra]

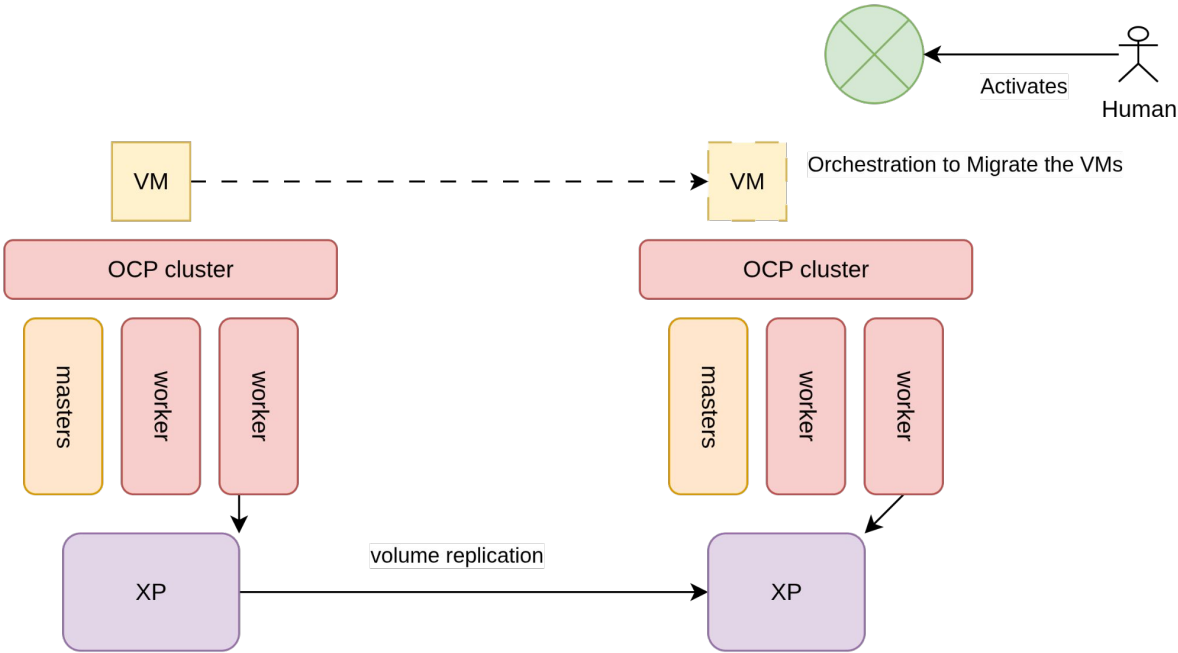
RTO	N/A
RPO	?? latency goal can be expressed in CR, but no minimum is documented



Component	Description
Volume Group Support	yes
Volume Replication Status management	Defined by CSI extension for 3PAR Defined in protection template for Nimble Note: it is not easy to reattach the VM on the passive site , complex automation required. This is to the point that this is not currently a viable solution
VM Management	Not provided
Recovery orchestration	Not provided

Regional DR [XP7/8]

RTO	N/A
RPO	??



Component	Description
Volume Group Support	It does not look like it is supported from the docs.
Volume Replication Status management	Define as a CSI extension . Replication Status can be managed in a fully declarative way.
VM Management	Not provided
Recovery orchestration	Not provided

Infinidat

Infinidat Storage Models

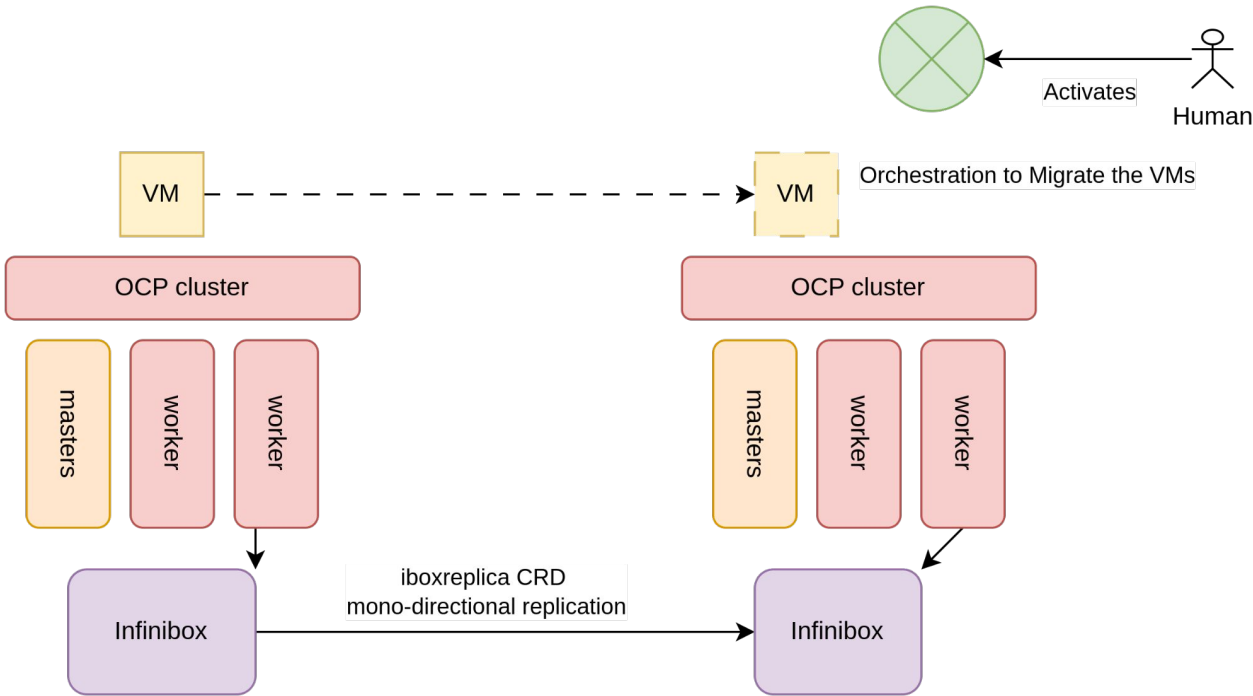
Model Family	Positioning	CSI Drivers	Supports Metro DR	Supports Regional DR
Infinibox		<u>csi</u>	No (the capability is not surfaced at the CSI-level)	yes

Metro DR



Regional DR

RTO	??
RPO	4s

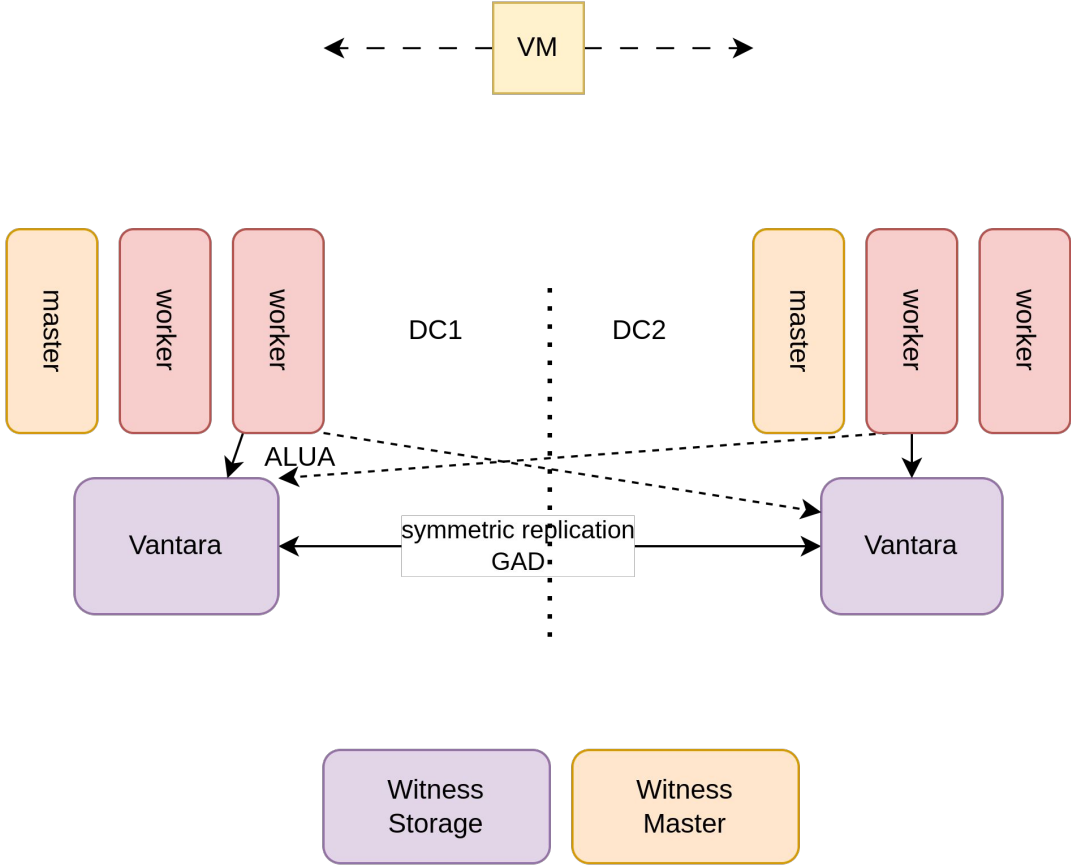


Component	Description
Volume Group Support	Yes, but not kube friendly (references infinibox entity instead of PVCs).
Volume Replication Status management	Defined by iboxreplica extension , replication status managed with kubectl. CR is not kube friendly.
VM Management	Not provided
Recovery orchestration	Not provided

Hitachi

RTO	~5 minutes
RPO	0

Metro DR



Symmetric replication for Vantara is called [GAD](#) (Global Active Device) and is documented [here](#).

GAD is supported in CSI with the concept of [stretched PVC](#).

Stretched PVC creates an active/passive multipath configuration (ALUA), favoring the local site.

```
defaults {
  find_multipaths yes
  user_friendly_names yes
}

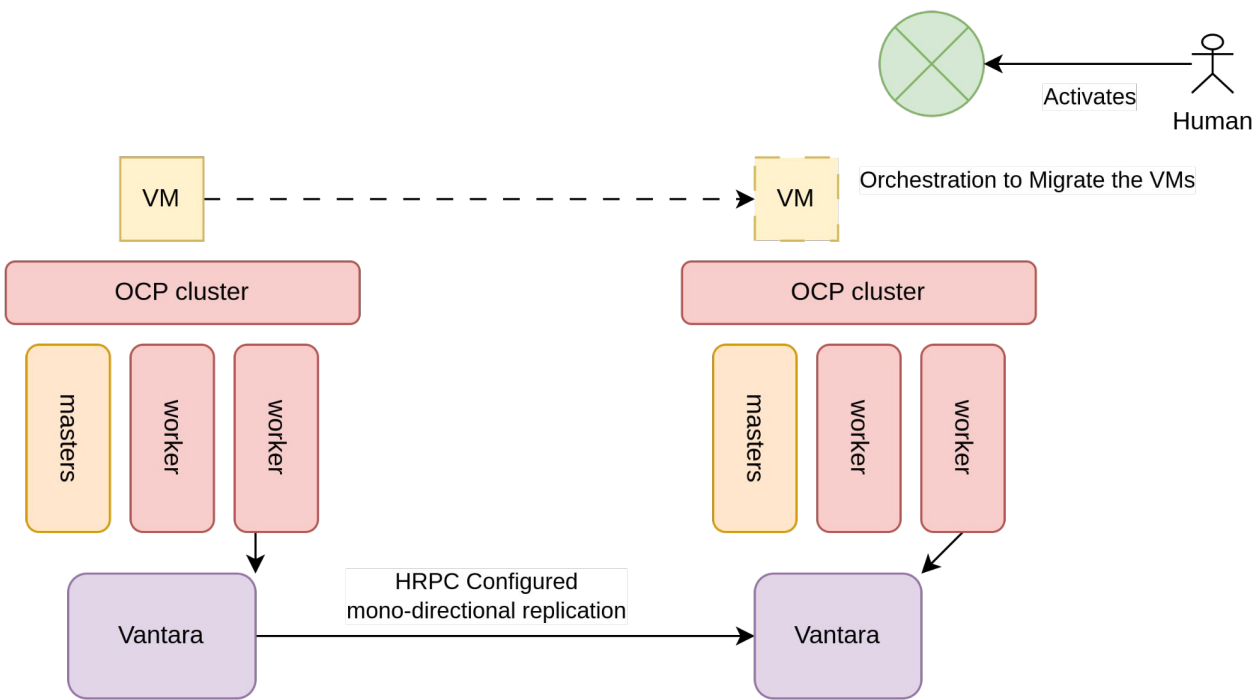
blacklist {
  devnode "rhd[a-z][0-9]*"
  devnode "sda5"
  devnode "sdb5"
}

devices {
  device {
    vendor "ENC"
    product "SWHETRIX"
    path_grouping_policy "group_by_prio"
    prio_weightedpaths
    prio_args "wm","r","0x50000072001a23ff:." 20 .".":"0x50000072001a2fff:." 15"
    path_selector "service-time 0"
  }
}
```



Regional DR

RTO	??
RPO	0?



Component	Description
Volume Group support	no
Volume Replication Status management	Defined by HRPC extension , replication status managed with kubectl. Only async is supported.
VM Management	Not provided
Recovery orchestration	Not provided

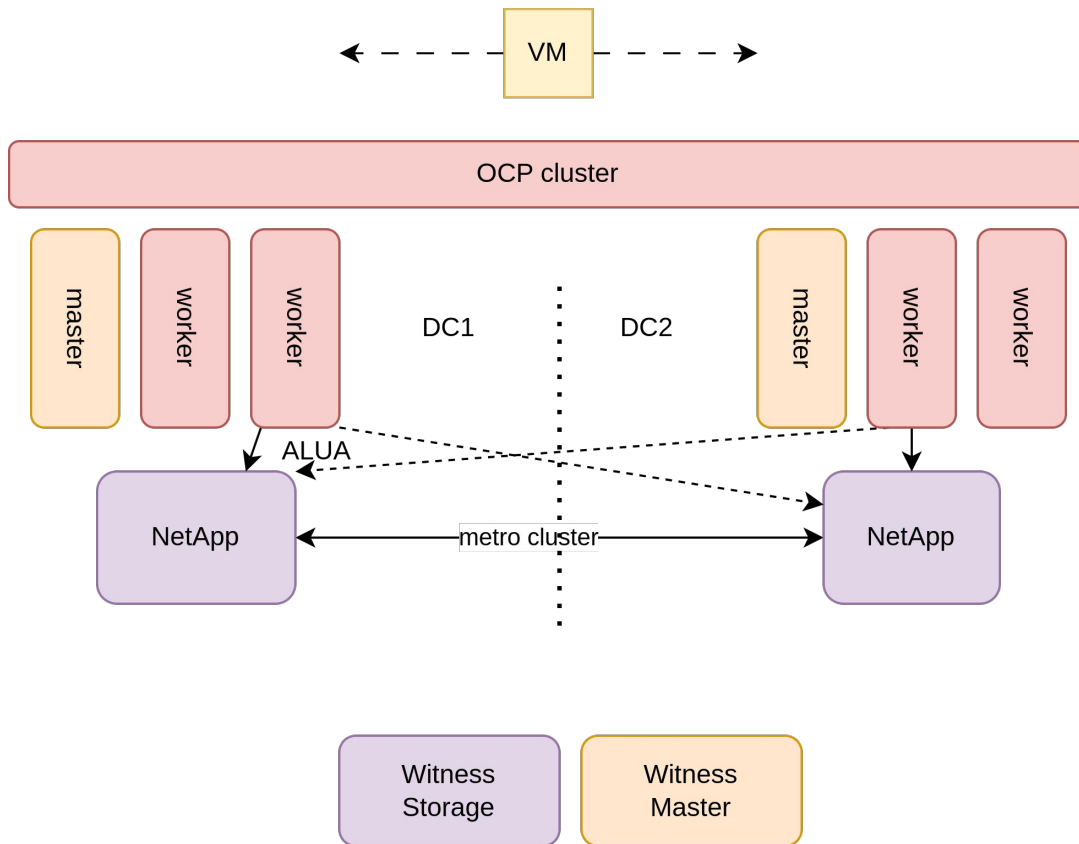
NetApp

NetApp Storage Models

Model Family	Positioning	CSI Drivers	Supports Metro DR	Supports Regional DR

RTO	~5 minutes
RPO	0

Metro DR



Symmetric replication for NetApp is called [MetroCluster Configuration \(MCC\)](#).

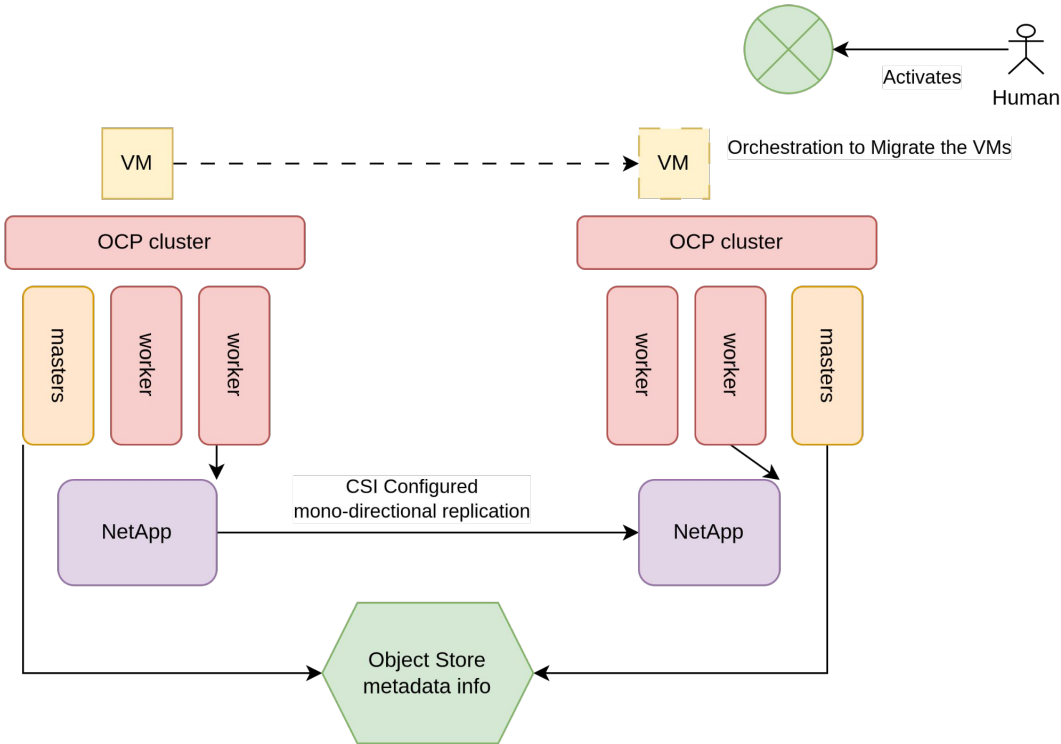
MetroCluster PVC volumes create active/passive multipath LUNs (ALUA).

With CSI topology read/write affinity to the closest storage arrays.

NetApp provides an optimized *multipathd.conf*.

Regional DR

RTO	N/A
RPO	Min 5 minutes



Replication technology based on the replication of delta snapshots. The VMs can be optionally freezed at the time of the snapshot (kubevirt vm-freeze).

Several use case are covered by *trident-protect*: failover, failback, manage replication direction.

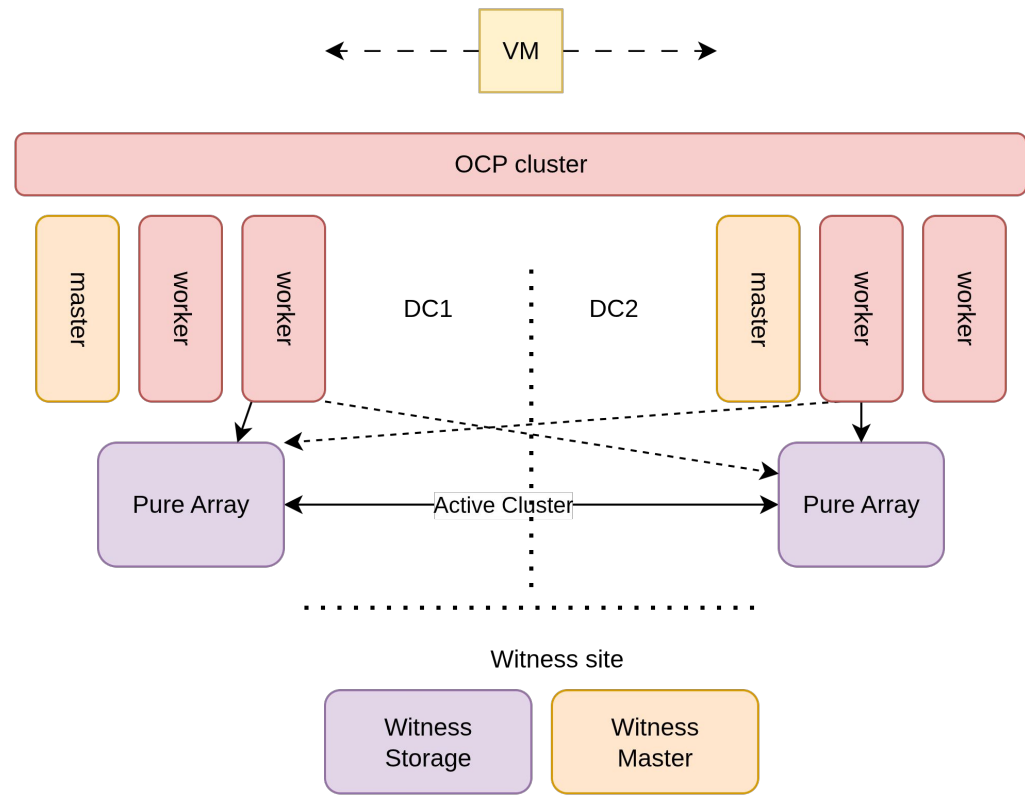
Component	Description
Volume group support	CSI Snapshot volume groups are used. But it's not a consistency group.
Volume Replication Status management	Defined by SNAP Mirror extension, replication status managed with <i>tridentctl-protect</i>
VM Management	Provided and managed via <i>tridentctl-protect</i> . Requires a bucket
Recovery orchestration	Not provided



Pure

RTO	~5 minutes
RPO	0

Metro DR



Symmetric replication is called [Active Cluster in Pure SAN Arrays](#) (see also [here](#)).

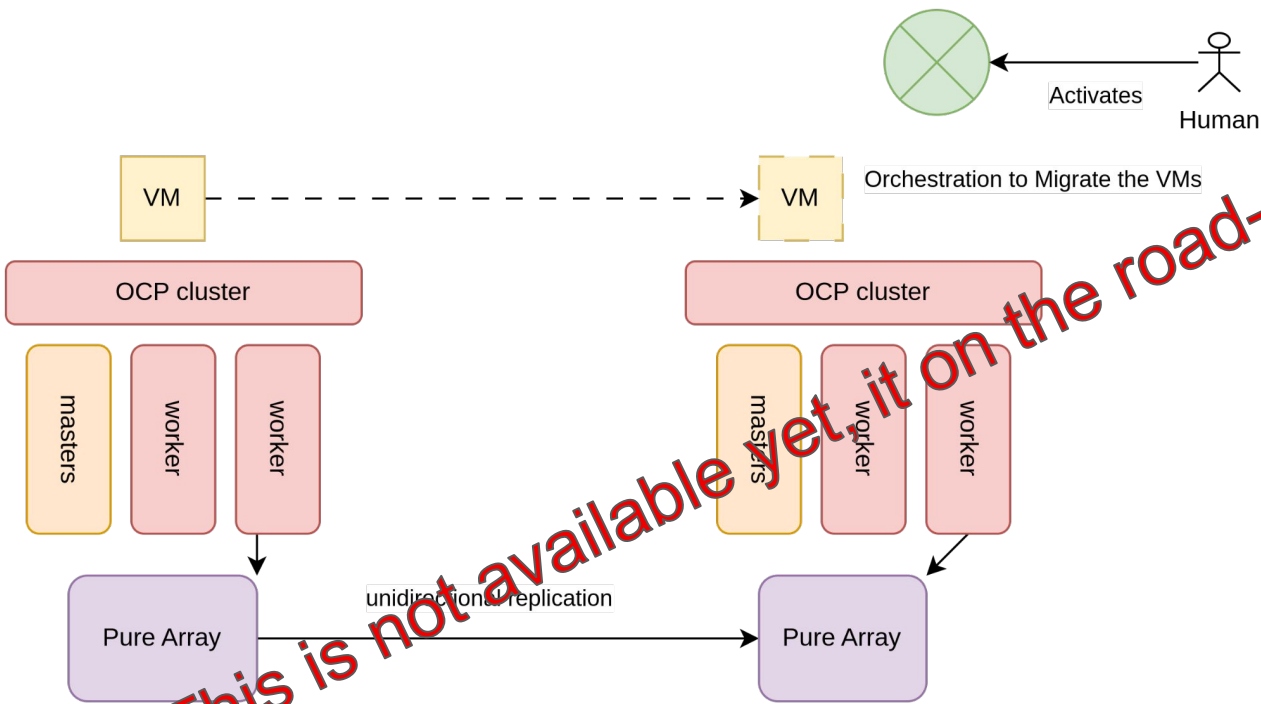
This can be used with [Flash Array Direct Access Volumes only](#).

Active Cluster PVC volumes create active/passive multipath LUNs (ALUA).

Regional DR

RTO	N/A
RPO	??

Unidirectional replication is available at the storage array level, but it is not currently bubbled up at the CSI level as a CSI extension.



Component	Description
Volume group support	N/A
Volume Replication Status management	N/A
VM Management	N/A
Recovery orchestration	N/A

IBM Storage Virtualize on FlashSystem Family

IBM Storage Virtualize Models

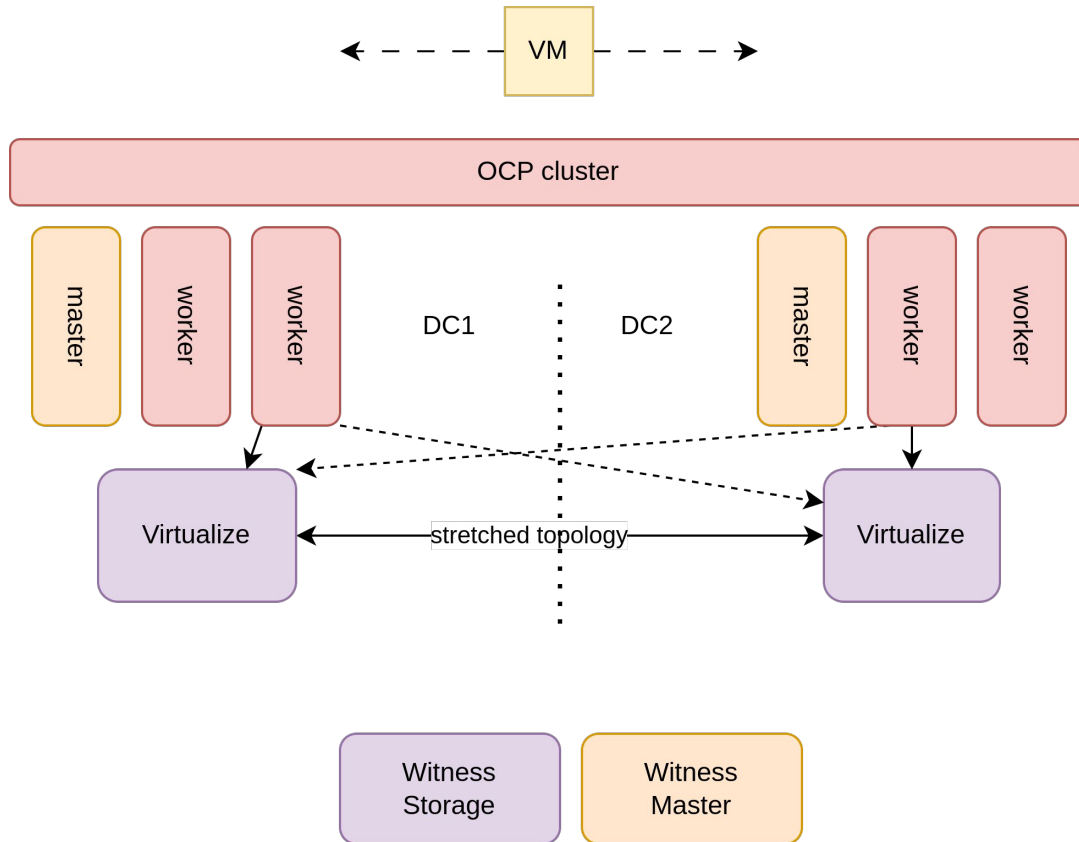
[Link to supported systems](#)

Model Family	Positioning	CSI Drivers	Supports Metro DR	Supports Regional DR
IBM Storage Flash System 5xxx	Entry level	<u>csi</u>	<u>yes</u>	<u>yes</u>
IBM Storage Flash System 7xxx	Mid range	<u>csi</u>	<u>yes</u>	<u>yes</u>
IBM Storage 9xxx	High range	<u>csi</u>	<u>yes</u>	<u>yes</u>
IBM SAN Volume Controller (SVC)	High range	<u>csi</u>	<u>yes</u>	<u>yes</u>

RTO	~5 minutes
RPO	0

Metro DR

A.k.a. Stretched or HA



Symmetric replication on Virtualize systems is called Hyperswap.

For SVC systems it is called or ESC (Enhanced Stretched Topology).

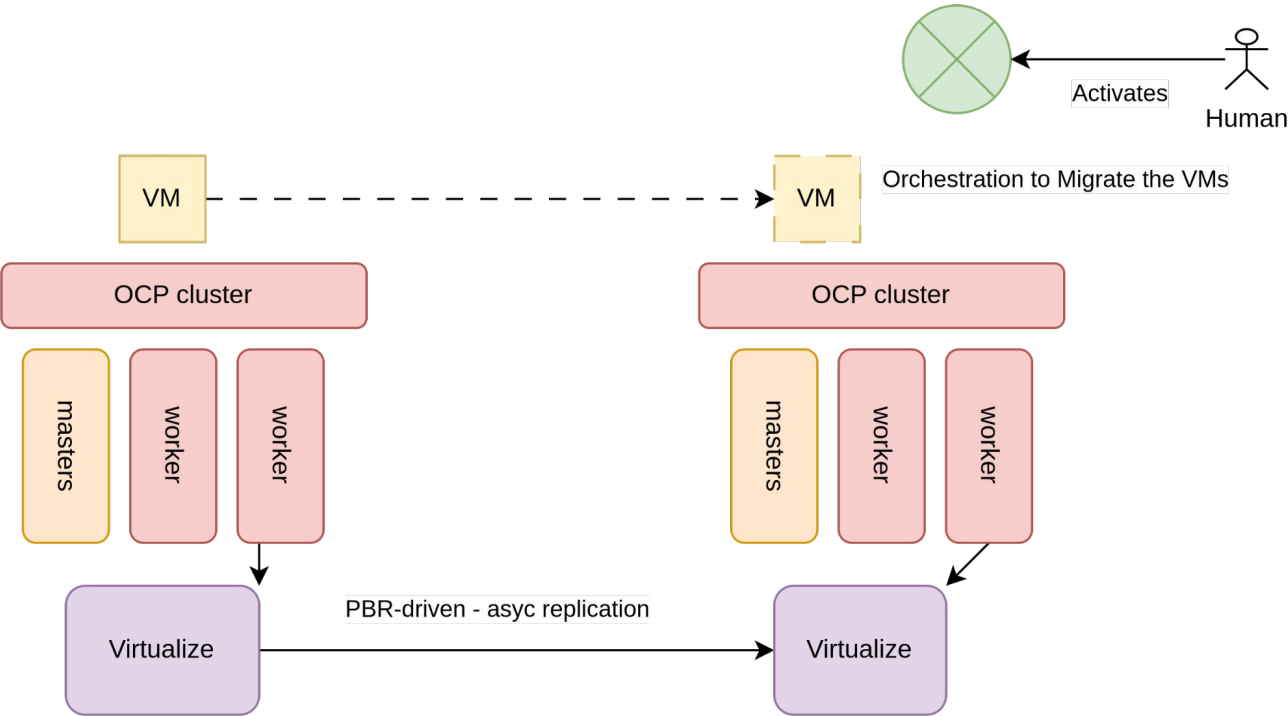
MetroCluster PVC volumes create active/passive multipath LUNs (ALUA).

With CSI topology read/write affinity to the closest storage arrays.

IBM provides an optimized *multipathd.conf*.

Regional DR

RTO	N/A
RPO	Min 5 minutes



Component	Description
Volume group support	Yes via custom CSI extension
Volume Replication Status management	Yes via VolumeReplication and VolumeReplicationOperator <i>behind the scene</i> PolicyBasedReplication is used (PBR)
VM Management	No, but ACM can be used for that with the same approach as for ODF(*)
Recovery orchestration	No.



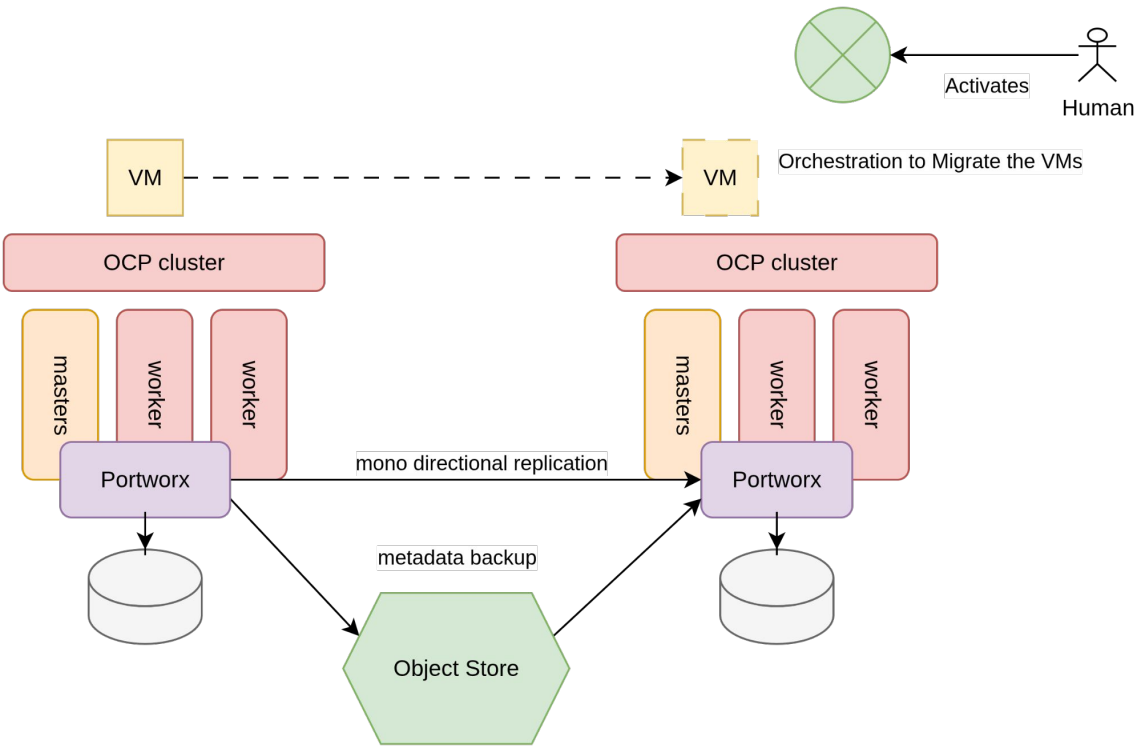
(*) if using ACM an approach needs to be found for ACM availability during a disaster.

SDS

Portworx

Regional DR

RTO	N/A
RPO	??



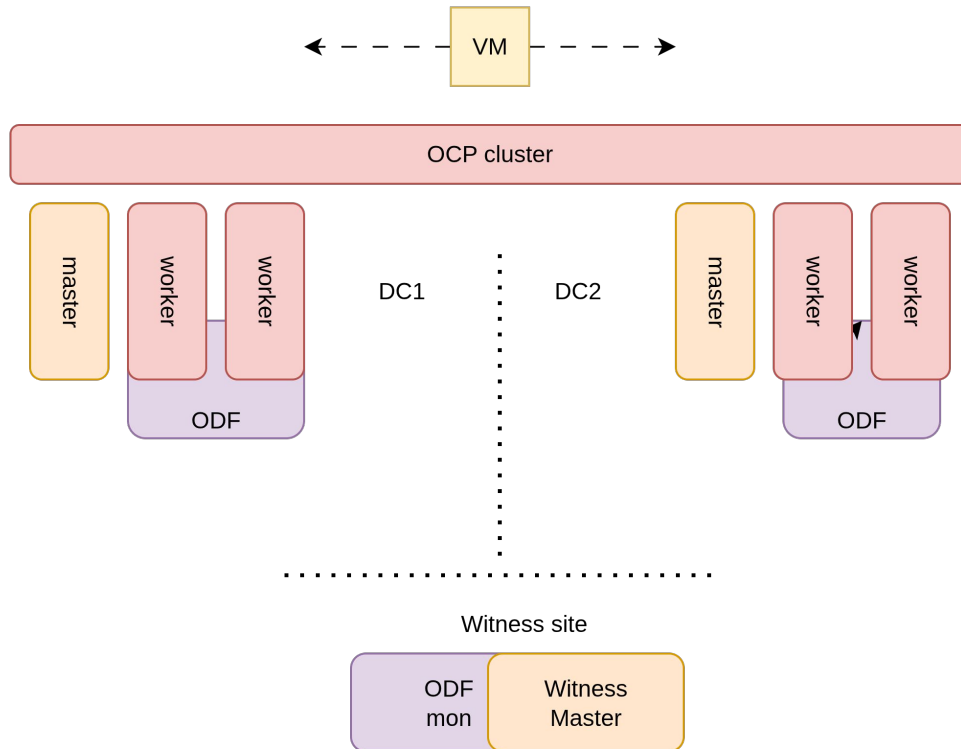
Component	Description
Volume group support	?? I think only at the namespace level
Volume Replication Status management	Defined by <u>migration schedule</u> extension, replication status managed with storkctl
VM Management	Provided and managed via storkctl
Recovery orchestration	Not provided



ODF / Fusion SDS

RTO	~5 minutes
RPO	0

Stretched OCP and ODF



OCP is stretched across two DC with a master node in witness site.

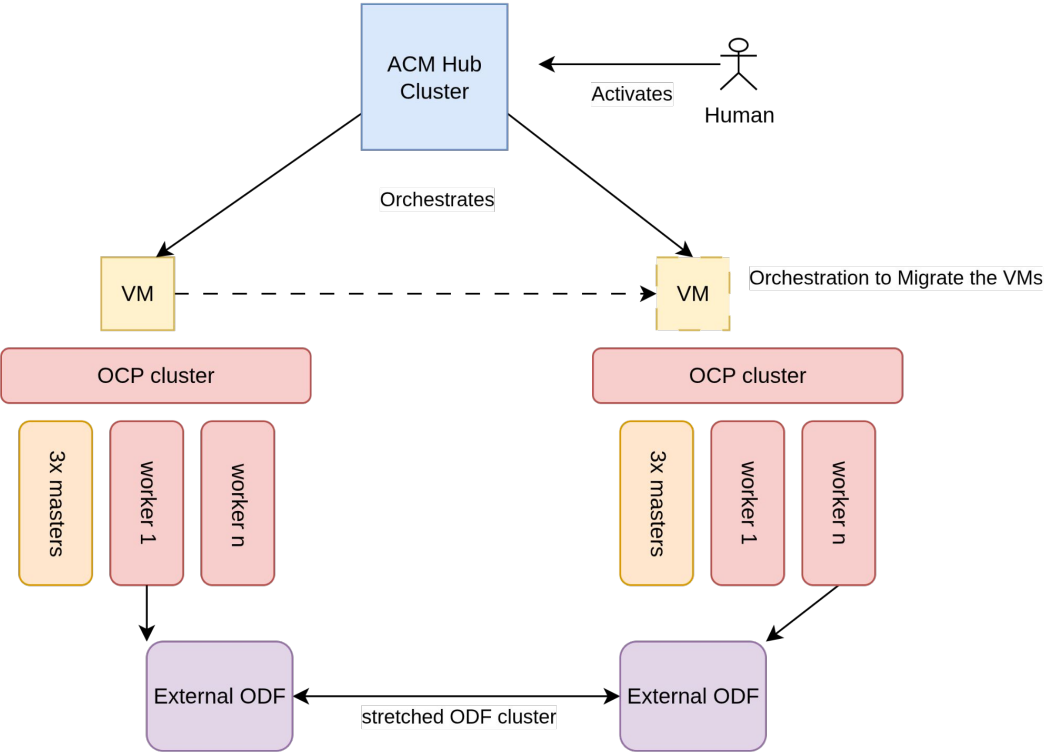
ODF can be also stretched and deployed on storage nodes. The quorum pod must be deployed on the witness master.

RTO	N/A
RPO	0

Metro DR

This architecture is dependent on ACM, which in turn is a SPO, making the design awkward. The recommendation is to have the active ACM on the passive DC.

The stretched external ODF, will likely require a quorum node (mon) in a witness site.

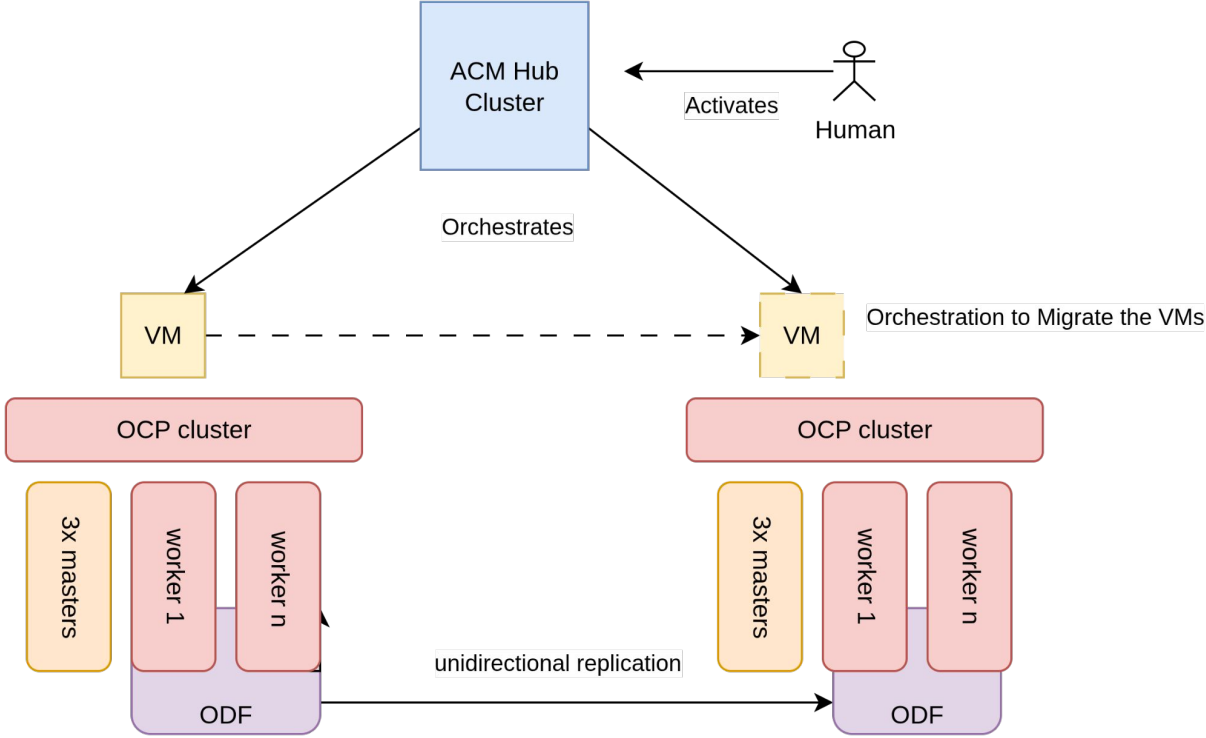


Component `	Description
Volume group support	Yes via VolumeGroup addon
Volume Replication Status management	Yes via VolumeReplication addon. This API is then implemented by the volume replication operator
VM Management	ACM can orchestrate the restart of the VMs with the RamenDR
Recovery orchestration	No.

Regional DR

RTO	N/A
RPO	>5m

This architecture is dependent on ACM, which in turn is a SPO, making the design awkward. The recommendation is to have the active ACM on the passive DC.



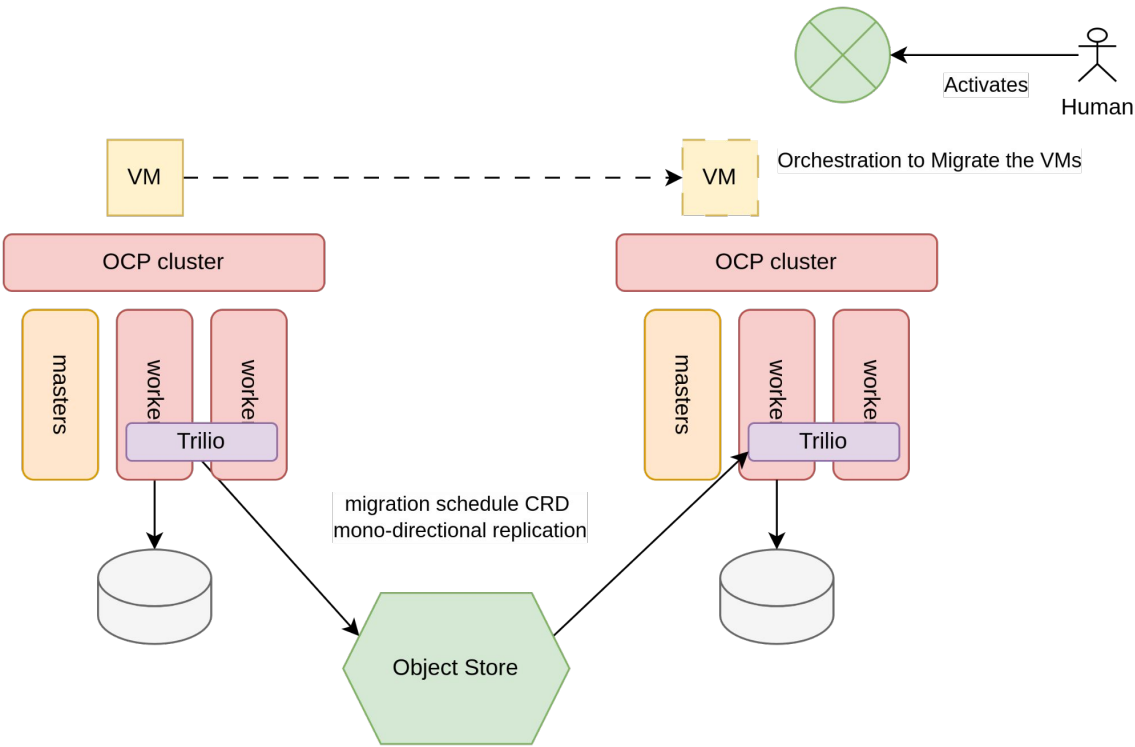
Component	Description
Volume group support	Yes via VolumeGroup addon
Volume Replication Status management	Yes via VolumeReplication addon. This API is then implemented by the volume replication operator
VM Management	ACM can orchestrate the restart of the VMs with the RamenDR
Recovery orchestration	No.

Backup and Restore tools (above the CSI line)

Trilio

Regional DR

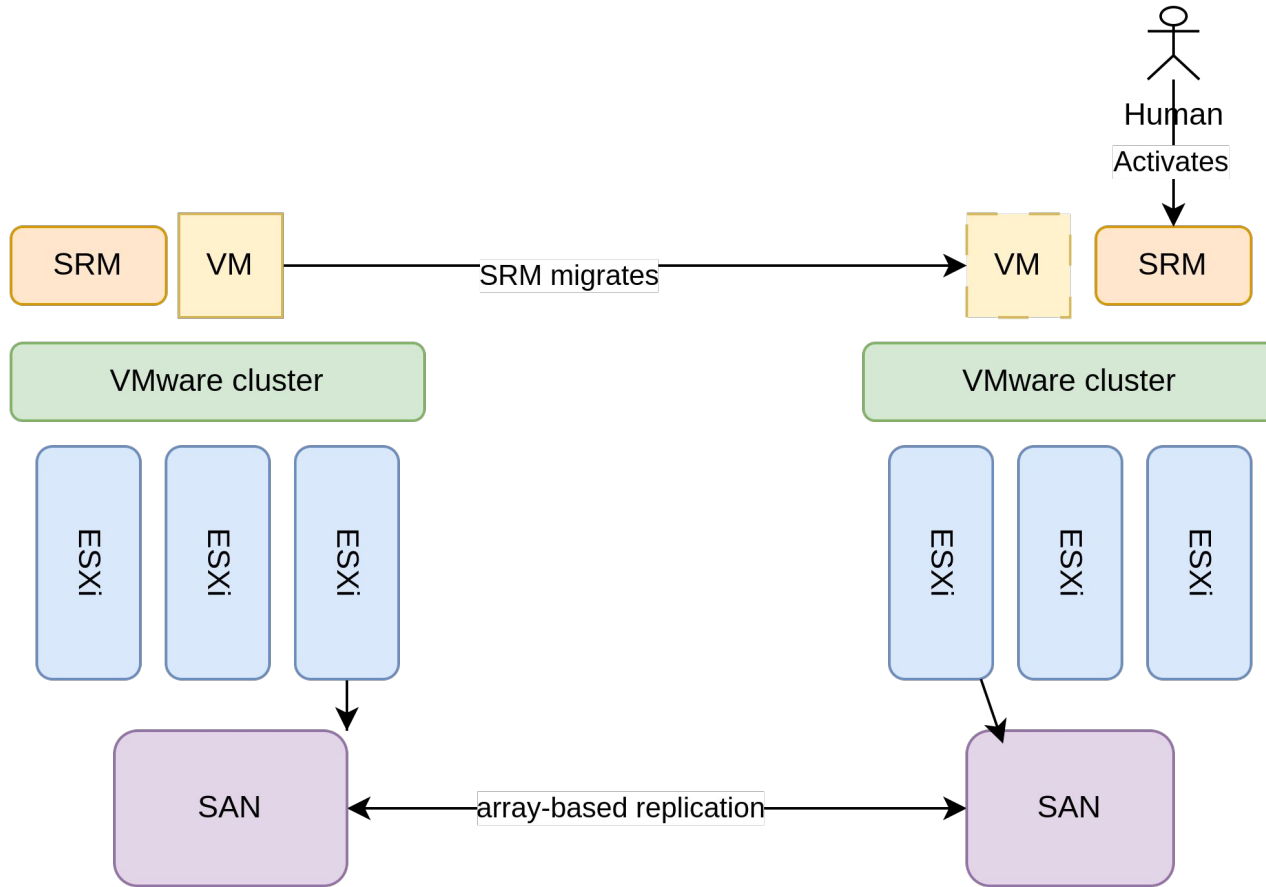
RTO	N/A
RPO	Min 15 minutes



Component	Description
Volume group support	Yes, based on VolumeSnapshotGroup
Volume Replication Status management	Replication define via the concept of <u>continuous restore</u>
VM Management	yes
Recovery orchestration	Not provided

Understanding VMware SRM and vMSC

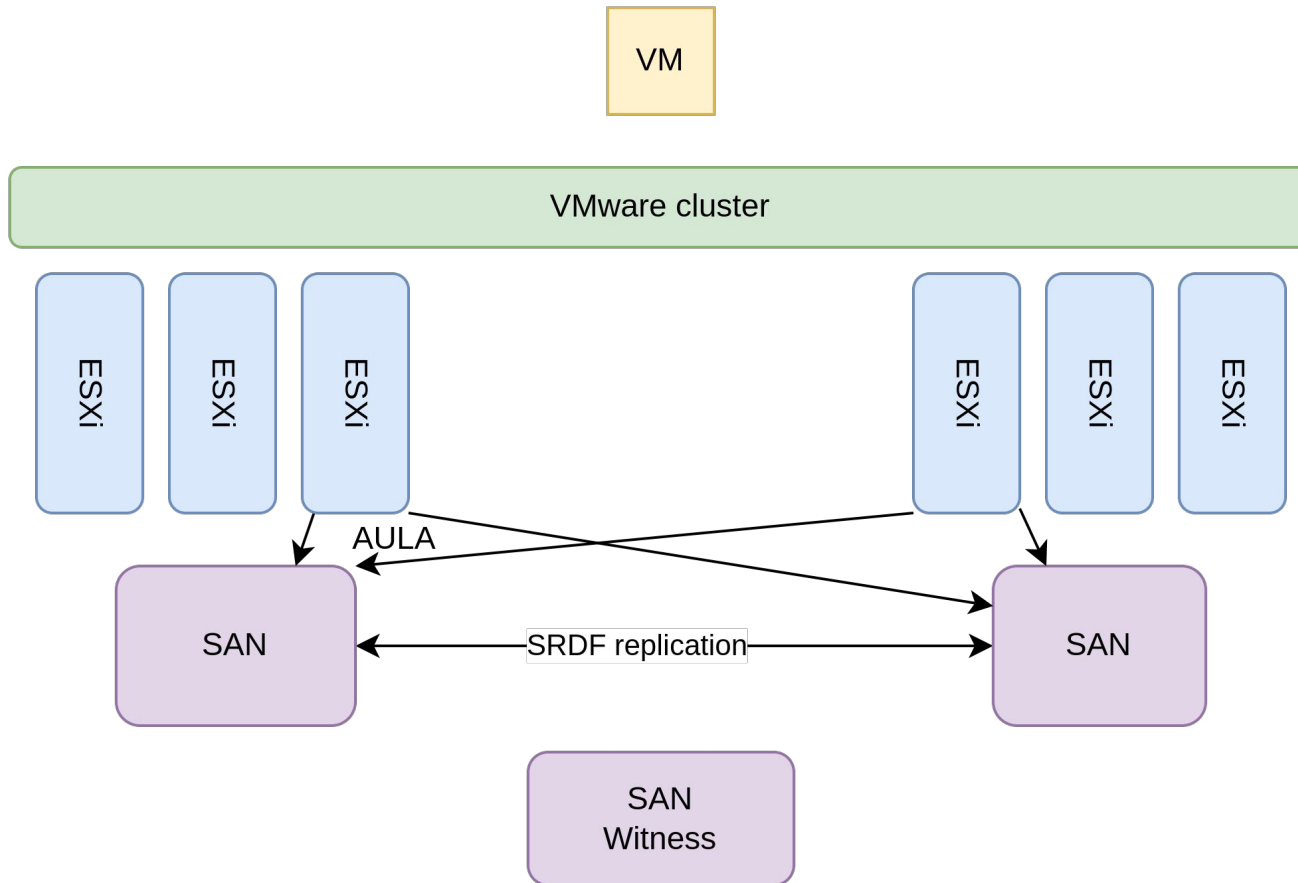
SRM



SRM sets up volume-based replication for selected machines.

- Sync/Async replication
- SAN array-based replication and SRA-based replication
- Any latency tolerated. It influences the RPO.
- When disaster hits, humans have to trigger the recovery process.
- Recovery procedure orderly restarts VMs on the other side.
- Symmetric deployment possible.

Symmetric storage replication+vMSC



Symmetric storage replication+vMSC sets up a stretched cluster across two metro DCs

- SAN must support symmetric storage replication. This requires a witness site and typically $\leq 5\text{ms}$ of latency.
- LUNs are setup with AULA, an active/passive protocol for iSCSI and FC
- The stretched SAN appears as a single logical SAN.
- When disaster hits, the system treats the situation as an HA event.

Thank you

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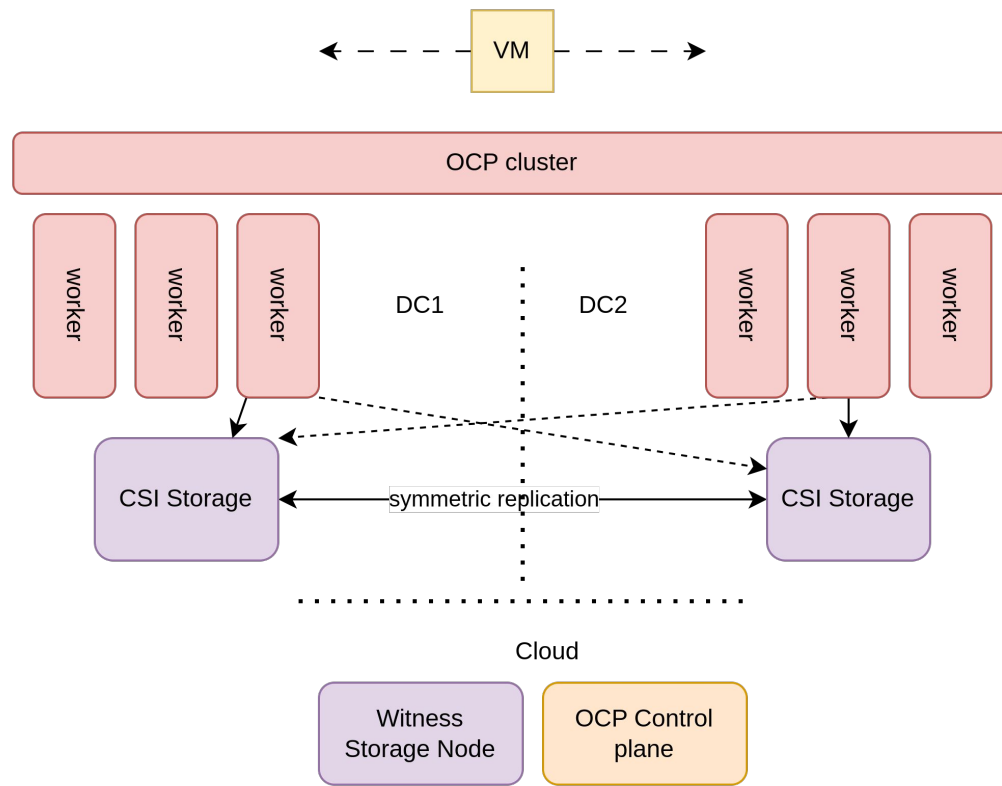
 youtube.com/user/RedHatVideos

 facebook.com/redhatinc

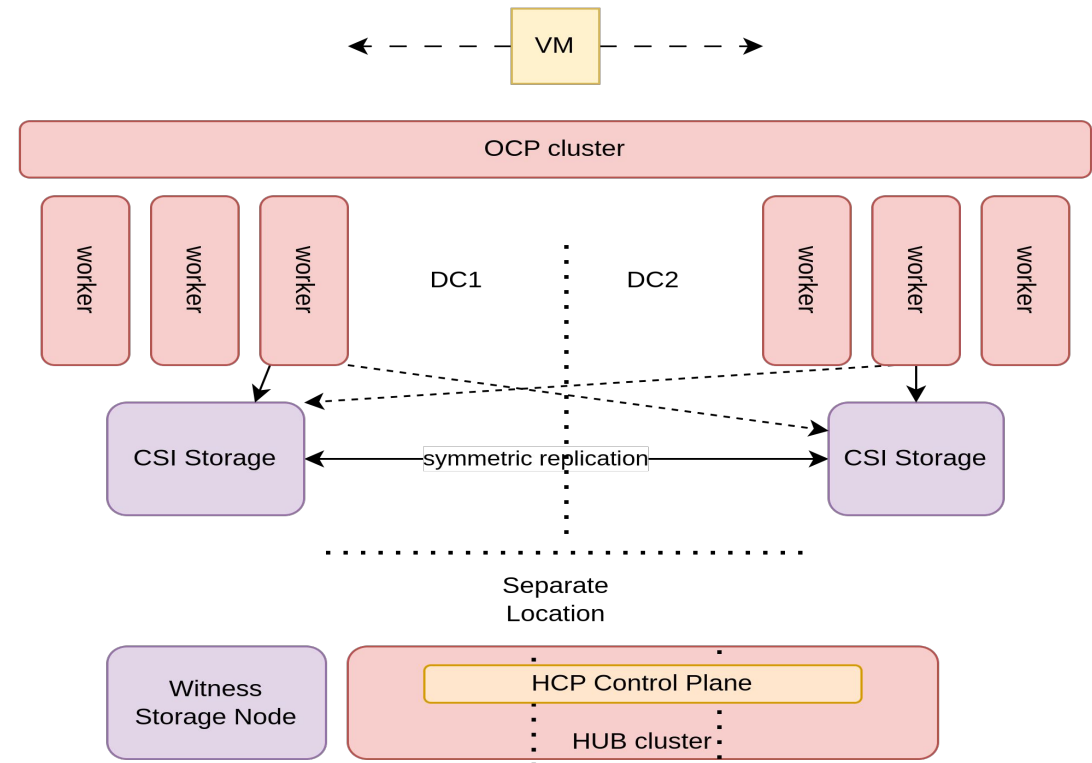
 twitter.com/RedHat

Metro DR -- Alternative architecture options

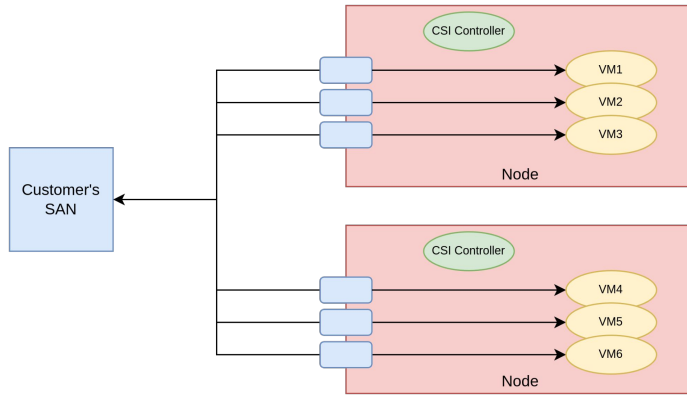
Control Plane in the cloud, nodes stretched across 2 DCs



HCP Control Plane in a cluster with 3 failure domains, nodes stretched across 2 DCs



Architectural Approaches for OpenShift Virt Storage when customer has a SAN



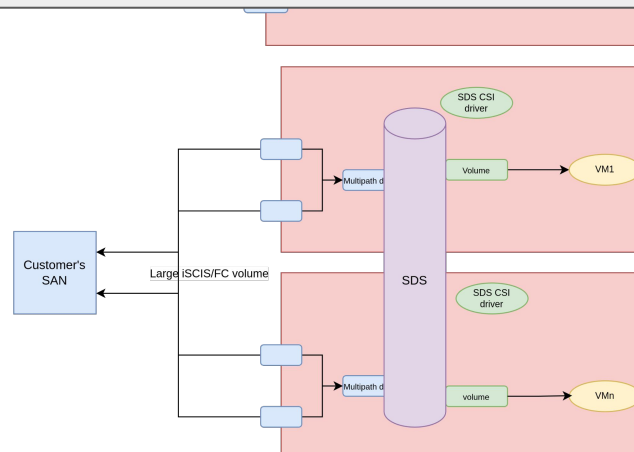
CSI Drivers: VMs directly attached to LUNs. One LUN per PVC

Pros:

- ▶ Kubernetes Native
- ▶ Can leverage the full SAN product capabilities

Cons:

- ▶ CSI drivers not always ready (buggy, untested at scale, not having the features needed for KubeVirt)
- ▶ Some SANs can only support a limited number of LUNs
- ▶ Some storage organizations do not want to give OCP credentials to provision LUNs. or may have practices that don't work with small/dynamic LUNs (LUN-centric practices)



SDS Overlay: sds storage on top of NAS. One LUN per storage node

Pros:

- ▶ Same behavior irrespective of the underlying SAN
- ▶ SDS solutions work typically well with KubeVirt

Cons:

- ▶ Additional licensing costs
- ▶ Additional latency
- ▶ Potential additional cost to upgrade networking to get storage over IP to have the same performance as FC.
- ▶ Write amplification: SDSs write the same data multiple times (ODF 3 times), increasing the storage needed on the SAN.

Status

Vendor	Status
Dell	Verified info. Requesting to create a reference implementation
HPE	Submitted info for verification.
Infinidat	Verified info. Requesting to create a reference implementation
Hitachi	Verified info. Requesting to create a reference implementation
NetApp	Verified info. Requesting to create a reference implementation
Pure	Submitted info for verification.
IBM Virtualize	Verified info. Requesting to create a reference implementation

1. Need vendor contact
2. Collecting information
3. Verify info with vendor
4. Waiting on vendor answers
5. Info validated
6. Reference implementation